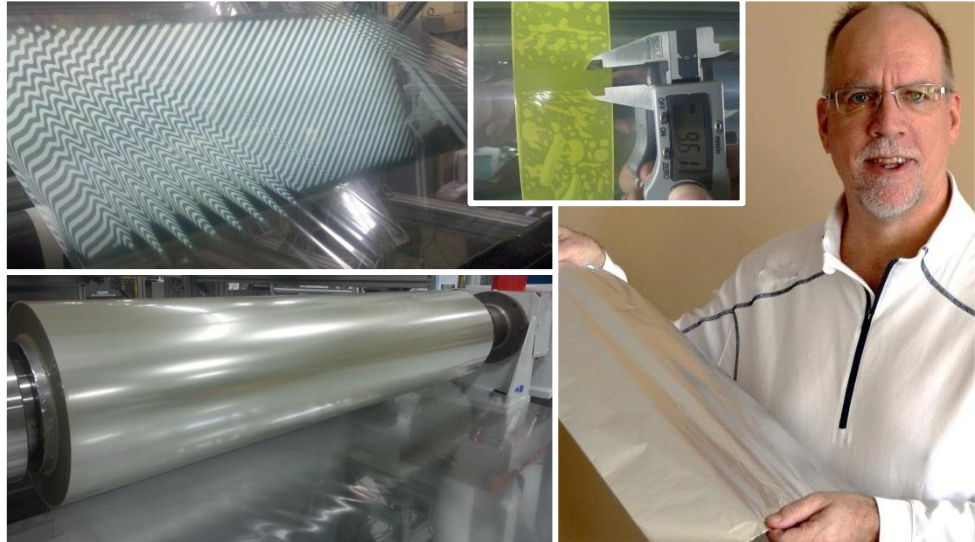




Web Handling Resources

Where to learn more about web handling

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116
tjwalker@tjwa.com www.webhandling.com 651.686.5400

WH Fundamentals

7: Web Handling Resources



Resources

Web Handling On-Line Resources

WH Fundamentals E-Book - Videos

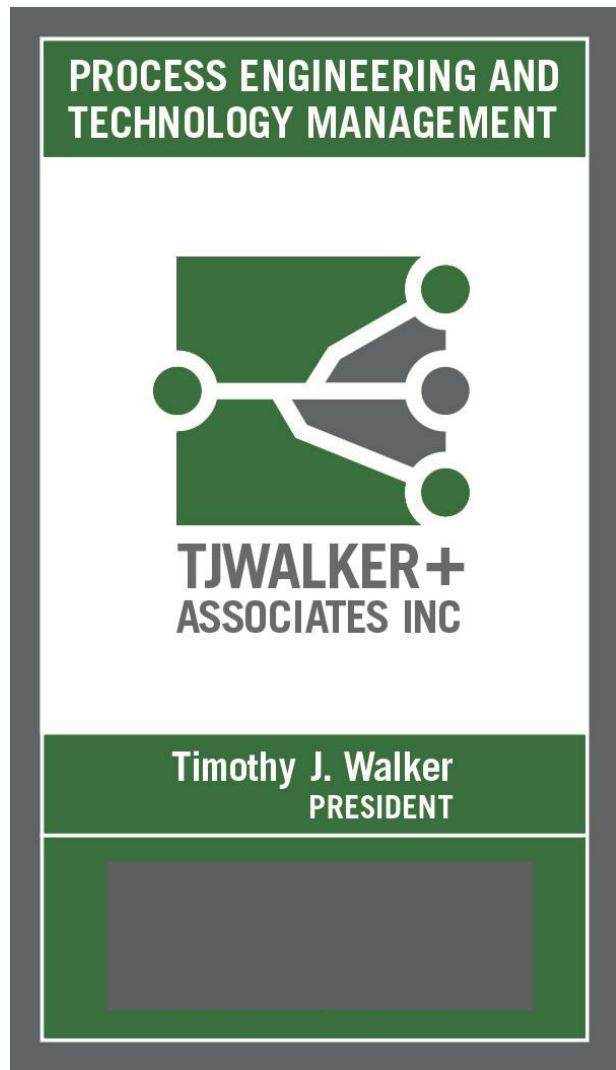
TopWeb Software

PFFC Web Lines Column Archive - WH Resources

+ Resources

- ✓ Where can I learn more about web handling?
- ✓ Where can I get help with web handling?

TJWalker+Associates, Inc.



- Web Handling Training
- Process Optimization
- Equipment Specification
- Vendor Selection (U.S.)

www.webhandling.com

tjwalker@tjwa.com

TJW Education & Experience

1983 *Univ. of Iowa, BS Mech. Engrg.*

1983-89 **3M Co. Polyester Film Division**

1989-99 **3M Company Web Handling Staff Consultant**

1994-99 **3M representative to Oklahoma St. Univ. Web Handling Research Center**

1995-99 **Led 300+ Web Process Design Reviews**

1997-99 **3M Corporate WH Technology Leader**

1999 **Univ. of Minn., MS Mgmt. of Technology**

1999-2041 **TJWalker + Associates Inc.
Process Engineering & Technology Management,
Specializing in Web Handling Processes**

TJWA Experience

Products / Materials

- ✓ Pressure Sensitive Adhesive Tapes
- ✓ Release Liners / films and papers
- ✓ Magnetic Products / PET, PEN
- ✓ Abrasives and Micro-Abrasives
- ✓ Flexible Electronics / Polyimide, Foils
- ✓ Flexible Solar / PET, Stainless Steel
- ✓ Flexible Packaging / Paper, Film, Foils
- ✓ Roofing Products
- ✓ Imaging Films / PET
- ✓ Reflective Products
- ✓ Batteries / Films and Foils
- ✓ Pharmaceutical
- ✓ Carbon Composites
- ✓ Paper-Like Films
- ✓ PTFE Films
- ✓ Specialty Fabrics



WebHandling.com HOME

Register for Free Web Handling E-Book Access & Newsletter: To register, click [HERE](#).

*For Information on upcoming Web Handling Workshops, click on **Workshops**.*

Welcome to Webhandling.com WH NEWS: More Workshops! I am excited to announce the offering of three more workshops from the webhandling.com team, all combining seminar with on-equipment demonstrations at Optimation's Media Conveyance Facility in Rochester, NY. Web Handling Workshop - Aug. 23-25, 2011 Web Tension and Speed Control Workshop - Sept. 20-21, 2011 Roller Performance Workshop - Sept. 22-23, 2011 Please join us for this practical, unique, and comprehensive web handling training.	TJWalker+Associates, Inc About - Consulting - Sponsored Training - Contact Web Handling Training (new!) Seminars - Workshops - Webinars Web Handling Services (new!) Nip System Solutions - Process Development Equipment Alignment - Process Audits Custom Equipment Web Handling On-Line Resources WH Fundamentals E-Book - Videos TopWeb Software PFFC Web Lines Column Archive - WH Resources
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for webhandling.com registered users

Web Handling Table of Contents

Tensioning Webs	Rollers + Traction	Guide-Spread-Wrinkle	Winding + Winders
- Introduction to Tensioning - Web Mechanics - Strain, The Key to Web Handling - Tensioning Basics - Tension Zones - Tension Control Basics - Torque Control Basics - Speed Control Basics - Draw Control Basics - Tension Feedback - Motors and Drives - Tension Variations - Bending and Twisting X - Baggy Webs / - Tension in Laminates - Registration	- Traction Basics / - Traction Limits X - Lubrication X - Roller Basics X - Alignment X - Nipped Roller Basics X - Rubber Covered Rollers X - Nip Pressure Variations X - Air Flotation and Drag Bars X	- Introduction to Web Tracking X - Tracking Rules / - Automatic Guide Basics X - Automatic Guide Mechanics X - Wrinkling Basics / - Wrinkling Remedies X - Spreading and Anti-Wrinkle Rollers X - More on Anti-Wrinkle Rollers X	- Introduction to Winding - The Winding Process / - Nip and Gap Rollers / - Wound Roll Pressure / - Winder Design / - Splicing and Roll Transfers X - Cinching and Telescoping / - More Winding Defects X - Winding and Thickness Variations X

TJWalker PFFC Web Lines Archive Links



WebHandling.com

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[TJWA Inc.](#)
[WH Training](#)
[WH Services](#)
[Online Resources](#)

PFFC Web Lines Column Archive by Topic

Web Lines Archive: Links to Paper, Film, and Foil Converter

I've written over 100 columns for the converting industry trade journal Paper, Film, and Foil Converter. All are archived at www.pffc-online.com in reverse chronological order.

Most people are searching for an answer, so I have organized the links to my PFFC columns by topic with description of each below.

Topics/Titles	Date, and Description
Web Properties	
The Long & the Short of Web Bending	Sep03: How much force is required to bend a web? How does the force to bend the web change with span length (the distance between rollers)?
Who Likes Film?	Jan05: Why do customers like products made with films? What are the advantages and disadvantages from a process engineering viewpoint?
Twisting: Twist & Shout	Jan07: How far can you twist a web before the center goes slack? Why is this common twisting specification not conservative enough?
Bending: Don't Get Bent Out of Shape	Feb07: How far can you bend a web until an edge goes slack? How much force does it take?
Traction	
Optimizing Traction: Goldilocks and the	Aug02: A high level of web-roller traction will prevent slip and scratch problems. A low level of web-roller traction may prevent

PFFC-Online.com



PFFC
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PFFC Buyers Guide

Access the industry's most complete Manufacturer & Supplier Directory

HOME » WHO LIKES FILM?
AUGUST 05, 2011

Surface Prep	Carton/Box	Coating/Laminating	Controls	Web/Roll Handling	Slitting	Printing	Flexpack	Label/Tape	Bag/Pouch	Die-Cutting	Trends/Management
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BROWSE BACK ISSUES

Who Likes Film?

Jan 1, 2005 12:00 AM, Timothy J. Walker, TJWalker & Assoc. Inc.

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Web Lines

Customers like films because they are cool. Films can be clear, thin, shiny, clean, and elastic. Product designers like films because they are functional. Films can improve a product's break strength, tear resistance, and gas barrier performance. Process people like films because...wait. Do process people like films?

Ask a process person whether he or she likes to run films or papers and many will answer "paper." The first run of a thin film on a line designed for paper will be fraught with waste—waste from scratching, winding, and wrinkling. Yikes! Maybe process people don't like films. But since customers and product designers like films

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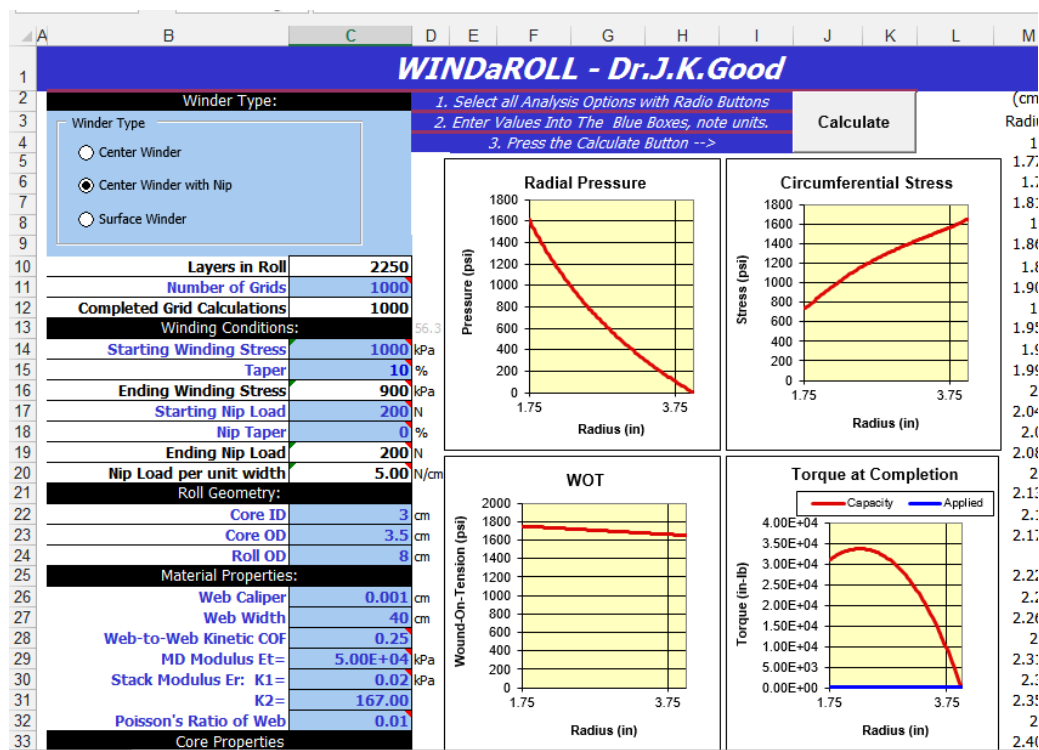
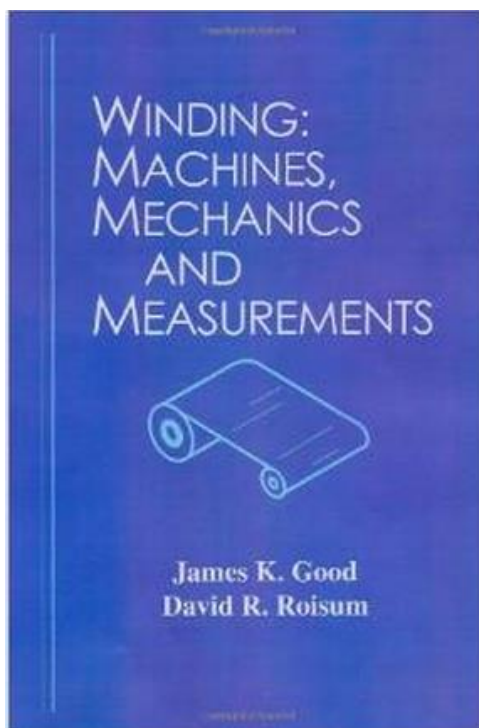
Wound Roll Stress Models

Winding: Machines, Mechanics, and Measurements

James K. Good and David R. Roisum

ISBN-13: 978-1932078695 (Listed at Amazon.com for \$175)

Book includes CD with WINDaROLL model (an Excel file running Visual Basic macro).



Web Handling (including Winding) Models

TopWeb 2.0 from Rheologic

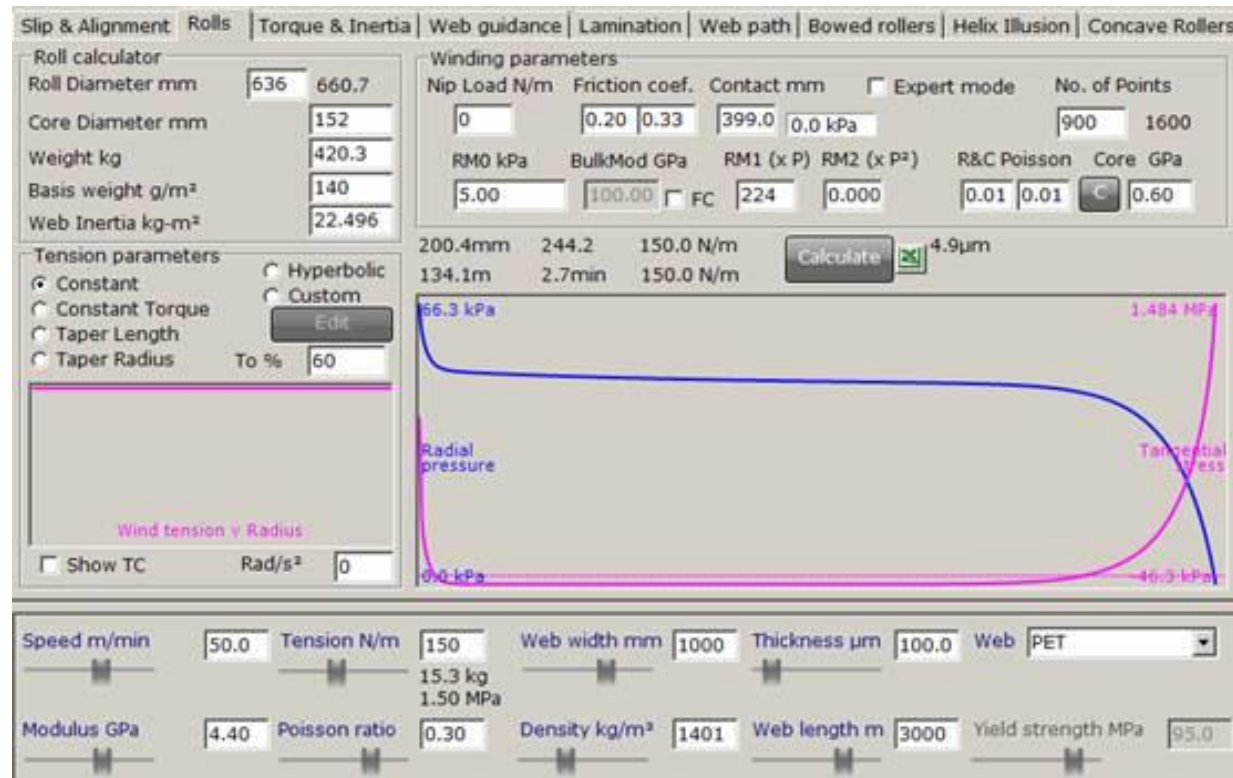


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Abbott Apps



Topic

Air gap between web and roller
 Heat sealing via Induction
 Heat sealing via Jaws
 Curl induced by lamination
 Roller Deflection under load
 Diffusion into wound rolls
 Dictionary of web terms
 Library of web handling
 Deformation of rubber roll
 Spreading of a web
 Control loop simulation
 Predicting web wrinkles
 Roll conversion productivity
 Calculating roll sizes
 Troubleshooting web wrinkles
 Choice of Web Tension

What

Air Lubrication Calculator
 Heat Seal Calculator - Induction
 Heat Seal Calculator - Jaws
 Laminate Curl Calculator (+THC)
 Roller Bending Calculator
 Roll Diffusion Calculator
 Roisum Dictionary
 Roisum Library
 Rubber Roller Calculator
 Spreader Type Selector
 Web Based Controller
 Wrinkle Predictor
 Winder Productivity Calculator
 Wound Roll Sizer
 Wrinkle Troubleshooter
 Web Tension Calculator

Web link

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Abbott Apps – Roller Bending Calculator

☐ Metric Units
 ☒ US Units
 ☒ Text
 Roller Bending Calculator

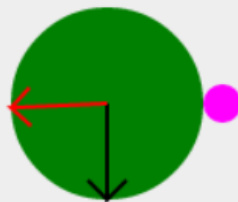


Web Load lb	0	Nip Load lb	2000
Web 1 ° ?	90	Nip Angle °	90
Web 2 °	270		
Roller OD in	6.5	Roller ID in	5
Face in	66	Brg-Brg in	72

☐ Steel
 ☒ Aluminum
 ☐ Low Composite
 ☐ Med. Composite
 ☐ High Composite
 ☐ Dead Shaft
 ☒ Live Shaft
 ☐ Cantilever
 ☐ Roll

Delta mil
 D/F Class

 Angle °
 React. lb





Optimation Technology, Inc.

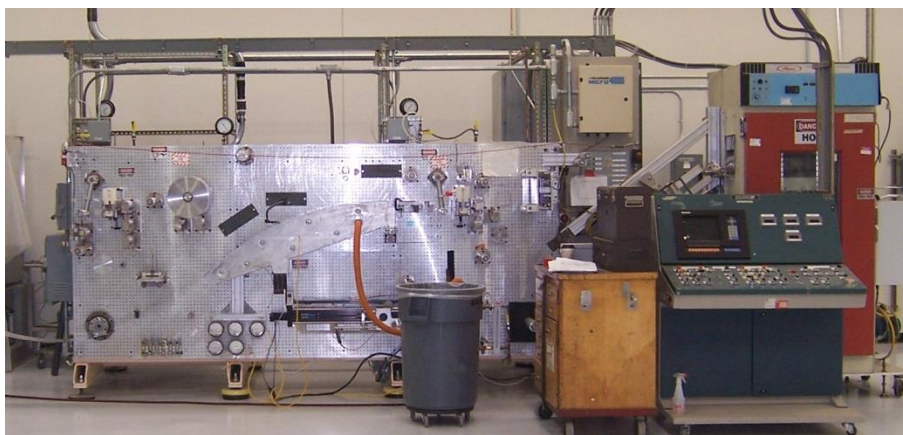
Media Conveyance Facility



Thin Web Rewinder



Laminator



Narrow Web Rewinder



Technical Training Center

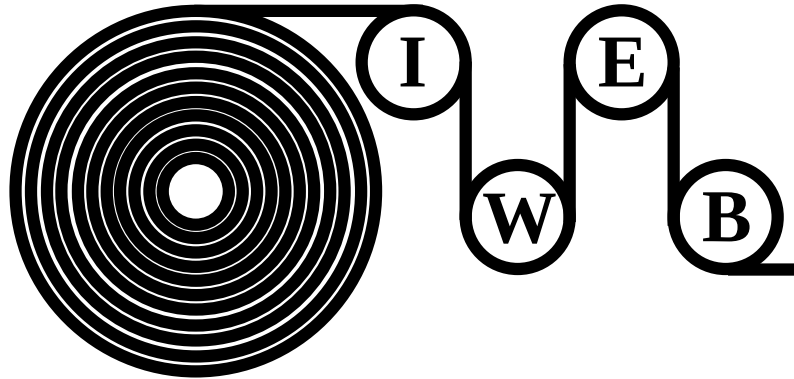
Web Handling Research Center – Oklahoma State University



- Mechanics of winding and unwinding
- Longitudinal dynamics and tension control
- Lateral dynamics and control; guiding and tracking
- Out-of-plane dynamics
- Wrinkling
- Air films between webs and rollers
- Aerodynamic effects in transport
- Measurement of tension, wound roll properties, and physical properties
- Precision lateral and longitudinal registration for roll-to-roll mfg.

**Founded in 1987.
Held 13th Int'l Conf.
on WH in 2015**

Web Handling Conference



*Sponsored by the
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Book Resources

The Ultimate Roll and Web Defect Troubleshooting Guide

WITH GLOSSARY

R. Duane Smith

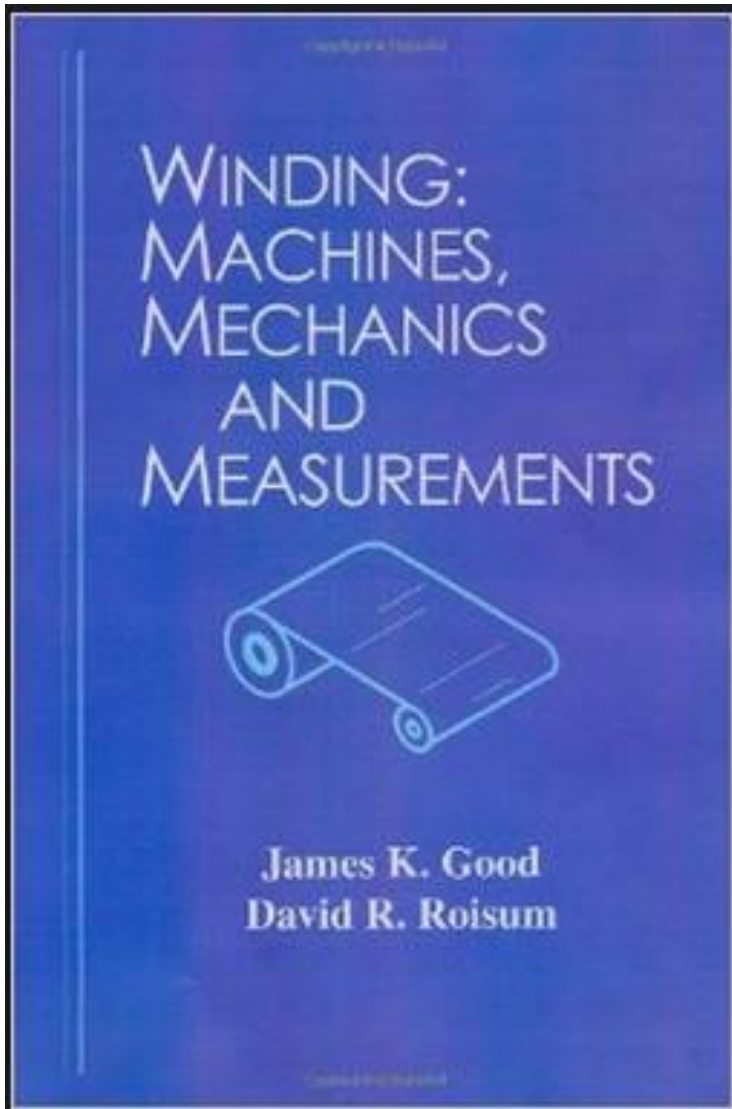


Alan Hadlock · Doug Howard · Bob LaPlant · Ted Lightfoot · Tony Lyons
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Each of the 224 defects has concise information which provides:

- A description of the defect
- An illustration of the defect
- The synonyms or “also known as” terms commonly used
- Cross references for these synonyms in index
- Common causes of the defect
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- Other sources of information published on the defect

Book Resources



Winding Machines: Mechanics and Measurement

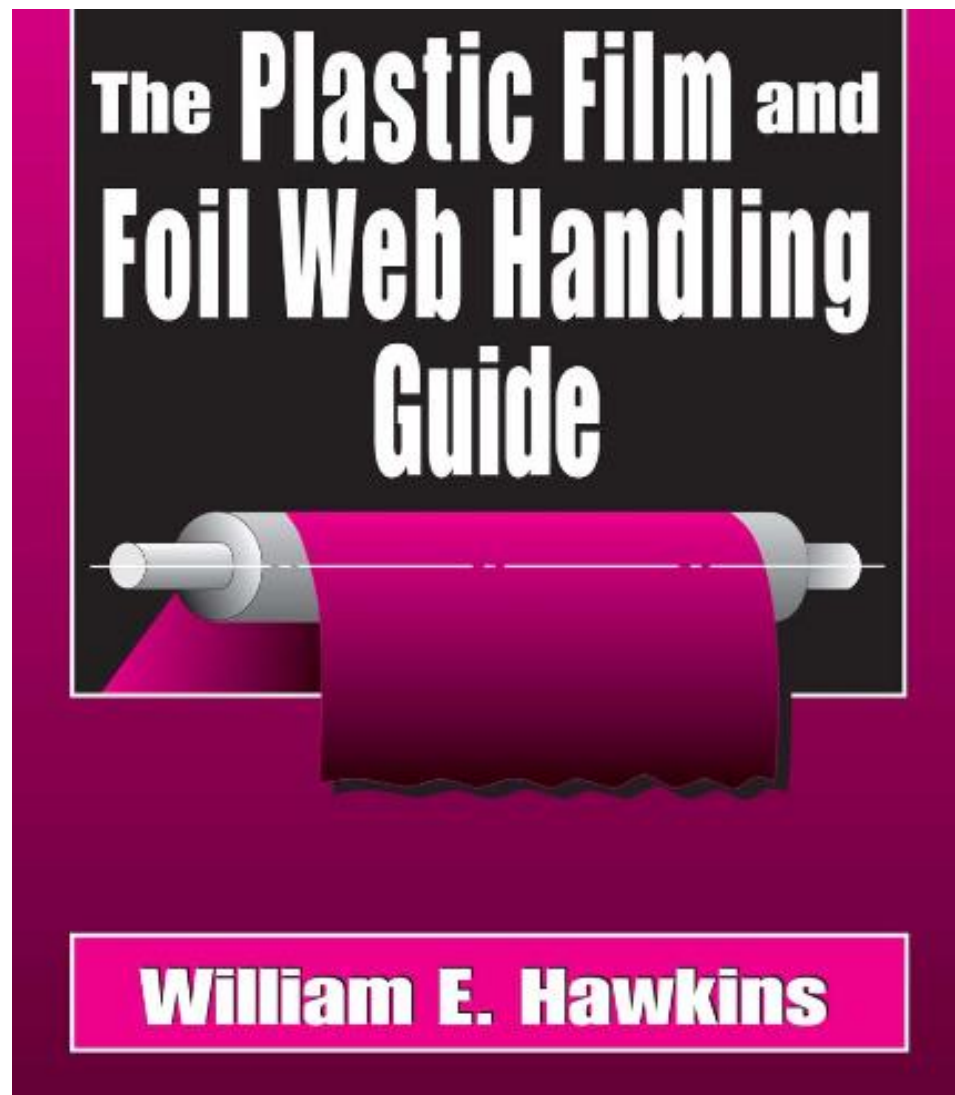
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Lancaster, Pennsylvania 17601 U.S.A.

Library of Congress Catalog Card No. 2006940785
ISBN: 978-1-932078-69-5

Includes CD with WindaRoll Model

Book Resources



Hawkins, William E.

The plastic film and foil web handling guide / William E. Hawkins.
p. cm.

Includes bibliographical references and index.

ISBN 1-58716-152-4

1. Papermaking machinery. 2. Winding machines. I. Title.